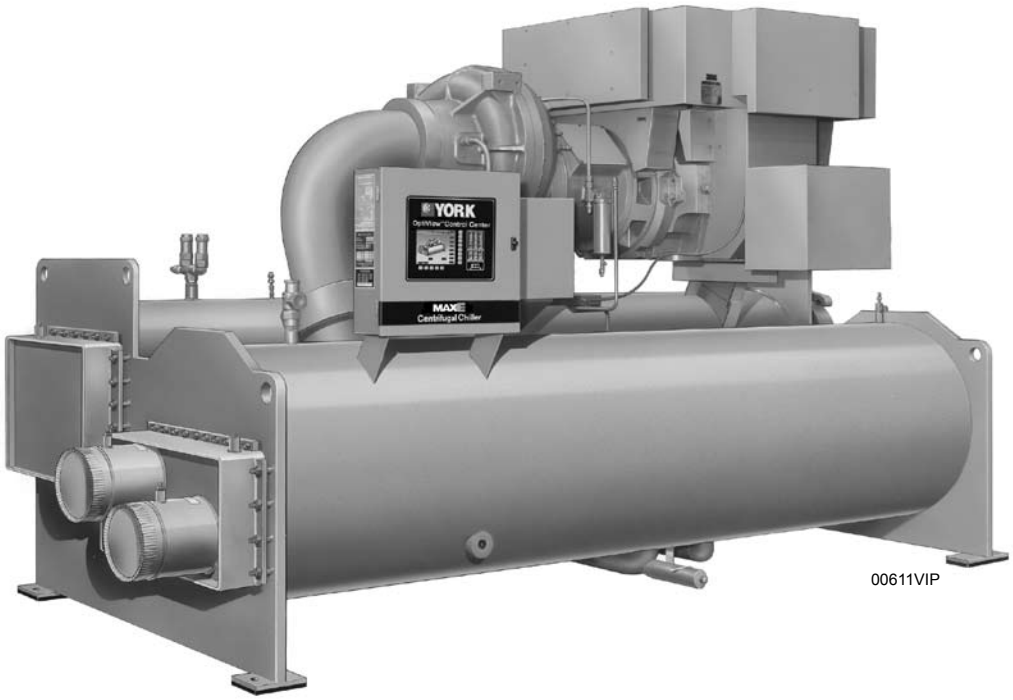
YORK BY JOHNSON CONTROLS	MAXE™ CENTRIFUGAL LIQUID CHILLERS	
OPERATING & MAINTENANCE	Supersedes: 160.75-O1	Form 160.75-O1

MODEL YK (STYLE G)
R-134a

WITH OPTIVIEW™ CONTROL CENTER FOR ELECTRO-MECHANICAL STARTER, SOLID STATE STARTER & VARIABLE SPEED DRIVE



00611VIP



LDO5842

IMPORTANT!
READ BEFORE PROCEEDING!
GENERAL SAFETY GUIDELINES

This equipment is a relatively complicated apparatus. During installation, operation, maintenance or service, individuals may be exposed to certain components or conditions including, but not limited to: refrigerants, oils, materials under pressure, rotating components, and both high and low voltage. Each of these items has the potential, if misused or handled improperly, to cause bodily injury or death. It is the obligation and responsibility of operating/service personnel to identify and recognize these inherent hazards, protect themselves, and proceed safely in completing their tasks. Failure to comply with any of these requirements could result in serious damage to the equipment and the property in which it is situated, as well as severe personal injury or death to themselves and people at the site.

This document is intended for use by owner-authorized operating/service personnel. It is expected that this individual possesses independent training that will enable them to perform their assigned tasks properly and safely. It is essential that, prior to performing any task on this equipment, this individual shall have read and understood this document and any referenced materials. This individual shall also be familiar with and comply with all applicable governmental standards and regulations pertaining to the task in question.

SAFETY SYMBOLS

The following symbols are used in this document to alert the reader to areas of potential hazard:



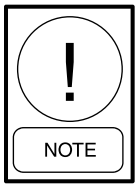
DANGER indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.



CAUTION identifies a hazard which could lead to damage to the machine, damage to other equipment and/or environmental pollution. Usually an instruction will be given, together with a brief explanation.



WARNING indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.



NOTE is used to highlight additional information which may be helpful to you.



External wiring, unless specified as an optional connection in the manufacturer's product line, is NOT to be connected inside the micro panel cabinet. Devices such as relays, switches, transducers and controls may NOT be installed inside the micro panel. NO external wiring is allowed to be run through the micro panel. All wiring must be in accordance with Johnson Controls published specifications and must be performed ONLY by qualified Johnson Controls personnel. Johnson Controls will not be responsible for damages/problems resulting from improper connections to the controls or application of improper control signals. Failure to follow this will void the manufacturer's warranty and cause serious damage to property or injury to persons.

CHANGEABILITY OF THIS DOCUMENT

In complying with Johnson Controls policy for continuous product improvement, the information contained in this document is subject to change without notice. While Johnson Controls makes no commitment to update or provide current information automatically to the manual owner, that information, if applicable, can be obtained by contacting the nearest Johnson Controls Service office.

It is the responsibility of operating/service personnel as to the applicability of these documents to the equipment in question. If there is any question in the mind of operating/service personnel as to the applicability of these documents, then, prior to working on the equipment, they should verify with the owner whether the equipment has been modified and if current literature is available.

REFERENCE INSTRUCTIONS

DESCRIPTION	FORM NO.
SOLID STATE STARTER – OPERATION & MAINTENANCE	160.00-O2
SOLID STATE STARTER – SERVICE INSTRUCTIONS	160.00-M4
VARIABLE SPEED DRIVE – OPERATION	160.00-O1
VARIABLE SPEED DRIVE – SERVICE INSTRUCTIONS	160.00-M1
FLOOR MOUNTED MV SSS - OPERATION	160.00-O5
UNIT MOUNTED MV SSS - OPERATION	160.00-O7
MEDIUM VOLTAGE SOLID STATE STARTER – SERVICE INSTRUCTIONS	160.00-M5
MEDIUM VOLTAGE VARIABLE SPEED DRIVE – SERVICE INSTRUCTIONS	160.00-M6
INSTALLATION	160.75-N1
OPTIVIEW™ CONTROL CENTER - SERVICE INSTRUCTIONS	160-54-M1
OPTIVIEW™ CONTROL CENTER - OPERATION AND MAINTENANCE	160.54-O1
WIRING DIAGRAM – FIELD CONNECTIONS	160.75-PW1
WIRING DIAGRAM – UNIT W/ REMOTE MV SSS	160.75-PW2
WIRING DIAGRAM – UNIT W/ MV VSD	160.75-PW3
WIRING DIAGRAM - FIELD CONTROL MODIFICATIONS	160.75-PW4
WIRING DIAGRAM - UNIT W/ MV SSS	160.75-PW5
WIRING DIAGRAM - UNIT W/ SSS, LV VSD W/ MODBUS, MV VSD	160.75-PW6
WIRING DIAGRAM - UNIT W/ LTC I/O BOARD W/ EM STARTER	160.75-PW7
WIRING DIAGRAM - UNIT W/ LTC I/O BOARD W/ MV VSD	160.75-PW8
RENEWAL PARTS – UNIT	160.75-RP1
RENEWAL PARTS – OPTIVIEW CONTROL CENTER	160.54-RP1

NOMENCLATURE

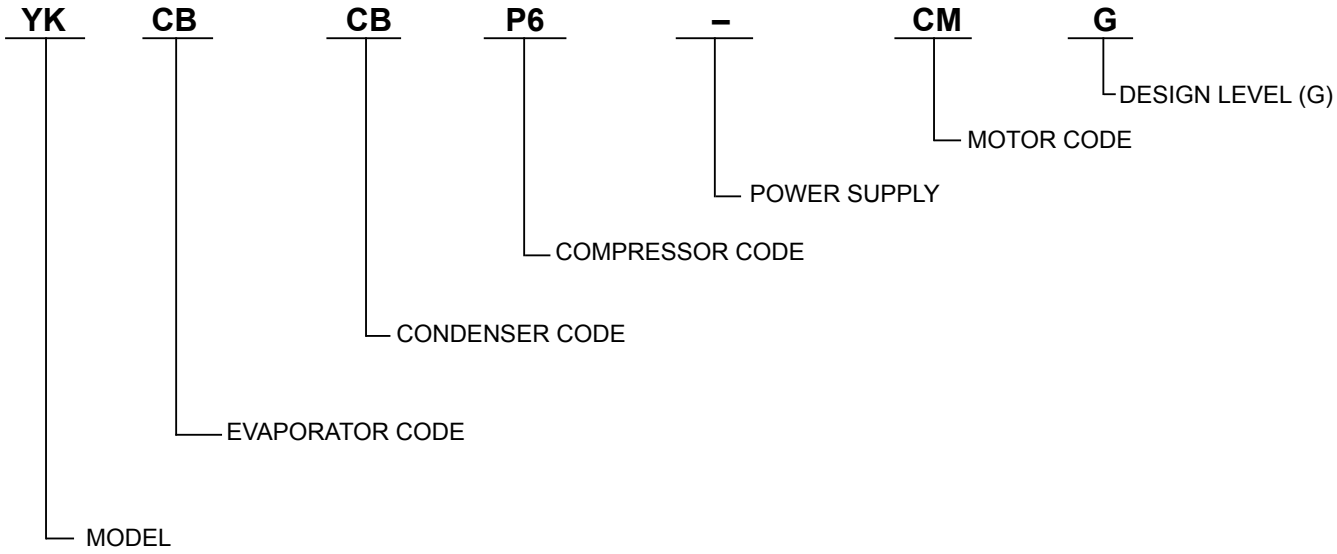


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SECTION 1
DESCRIPTION OF SYSTEM AND FUNDAMENTALS OF OPERATION

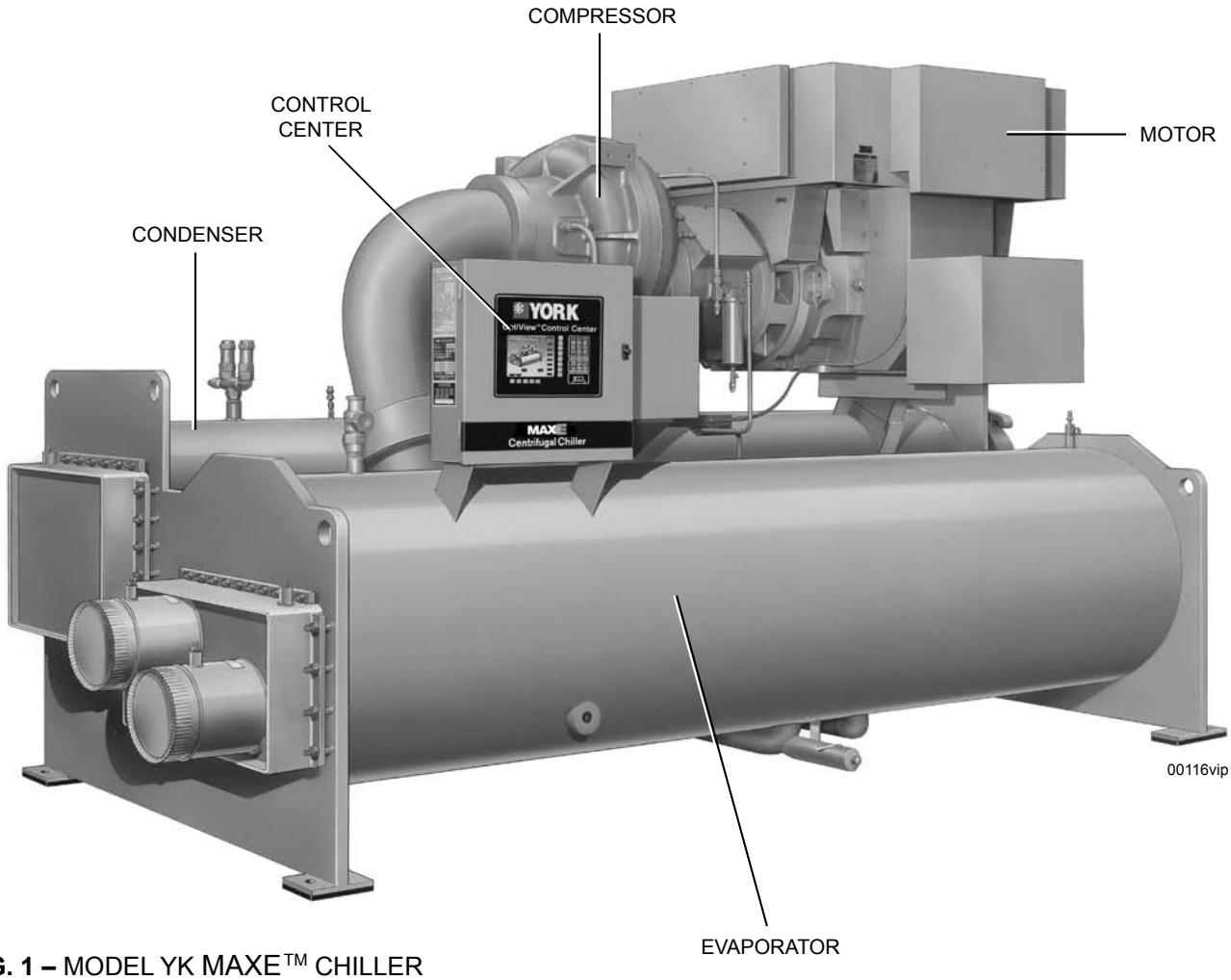


FIG. 1 – MODEL YK MAXE™ CHILLER

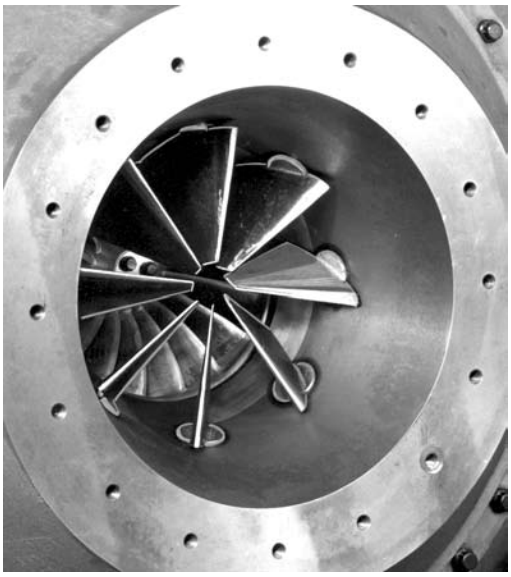
SYSTEM OPERATION DESCRIPTION (SEE FIG. 2)

The YORK Model YK MaxE™ Chiller is commonly applied to large air conditioning systems, but may be used on other applications. The chiller consists of an open motor mounted to a compressor (with integral speed increasing gears), condenser, evaporator and variable flow control.

The chiller is controlled by a modern state of the art Microcomputer Control Center that monitors its operation. The Control Center is programmed by the operator to suit job specifications. Automatic timed start-ups and shutdowns are also programmable to suit nighttime, weekends, and holidays. The operating status, temperatures, pressures, and other information pertinent to operation of the chiller are automatically displayed and read on a graphic display. Other displays can be observed by pressing the keys as labeled on the Control Center. The chiller with the OptiView Control Center is compatible with an electro-mechanical starter, YORK Solid State Starter (optional), or Variable Speed Drive (optional).

In operation, a liquid (water or brine to be chilled) flows through the evaporator, where boiling refrigerant absorbs heat from the liquid. The chilled liquid is then piped to fan coil units or other air conditioning terminal units, where it flows through finned coils, absorbing heat from the air. The warmed liquid is then returned to the chiller to complete the chilled liquid circuit.

The refrigerant vapor, which is produced by the boiling action in the evaporator, flows to the compressor where the rotating impeller increases its pressure and temperature and discharges it into the condenser. Water flowing through the condenser tubes absorbs heat from the refrigerant vapor, causing it to condense. The condenser water is supplied to the chiller from an external



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DETAIL A – COMPRESSOR PREROTATION VANES

source, usually a cooling tower. The condensed refrigerant drains from the condenser into the liquid return line, where the variable orifice meters the flow of liquid refrigerant to the evaporator to complete the refrigerant circuit.

The major components of a chiller are selected to handle the refrigerant, which would be evaporated at full load design conditions. However, most systems will be called upon to deliver full load capacity for only a relatively small part of the time the unit is in operation.

CAPACITY CONTROL

The major components of a chiller are selected for full load capacities, therefore capacity must be controlled to maintain a constant chilled liquid temperature leaving the evaporator. Prerotation vanes (PRV), located at the entrance to the compressor impeller, compensate for variation in load (See Detail A).

The position of these vanes is automatically controlled through a lever arm attached to an electric motor located outside the compressor housing. The automatic adjustment of the vane position in effect provides the performance of many different compressors to match various load conditions from full load with vanes wide open to minimum load with vanes completely closed.

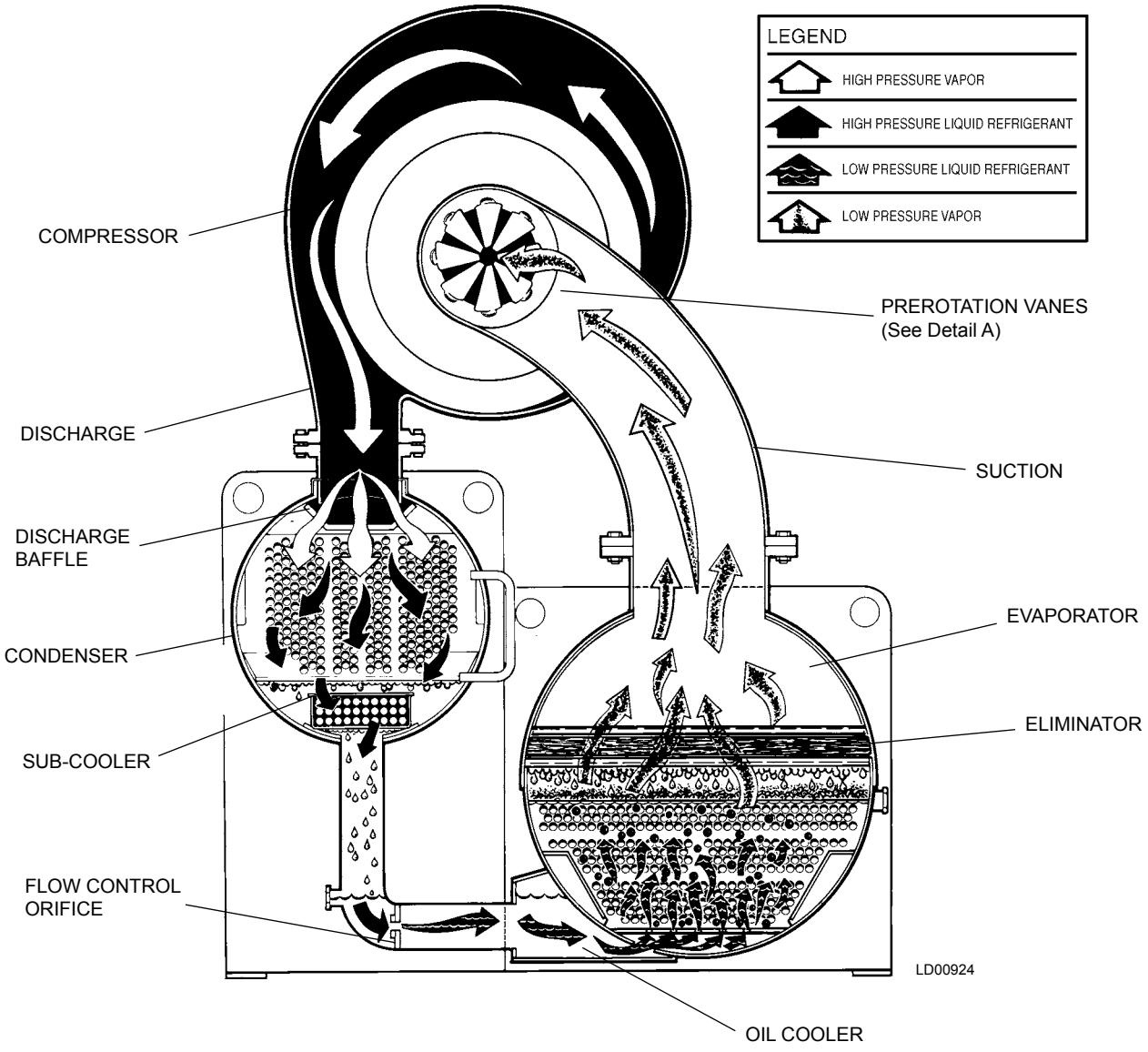


FIG. 2 – REFRIGERANT FLOW-THRU CHILLER

SECTION 2
SYSTEM OPERATING PROCEDURES

OIL HEATERS

If the oil heater is de-energized during a shutdown period, it must be energized for 12 hours prior to starting compressor, or remove all oil and recharge compressor with new oil. (See "Oil Charging Procedure", page 21.)

OIL HEATER OPERATION

The oil heater operation is controlled by the OptiView™ Control Center. The heater is turned on and off to maintain the oil temperature differential to a value 50°F (27.8°C) above the condenser saturation temperature. This target value is maintained by the control panel.

If the target value is greater than 160°F (71°C), the target defaults to 160°F (71°C). If the target value is less than 110°F (43.3°C), it defaults to 110°F (43.3°C).

To prevent overheating of the oil in the event of a control center component failure, the oil heater thermostat (1HTR) is set to open at 180°F (82°C).

CHECKING THE OIL LEVEL IN THE OIL RESERVOIR

Proper operating oil level – During operation, the oil level should fall to the "Operating Range" identified on the vertical oil level indicator label (See Figure 3).

- If the oil level during operation is in the "Over Full" region of the oil level indicator, oil should be removed from the oil reservoir, This reduces the oil level to the "Operating Range".
- If the oil level during operation is in the "Low Oil" region of the oil level indicator, oil should be added to the oil reservoir. (See "Oil Charging Procedure", page 21)



FIG. 3 – OIL LEVEL INDICATOR

LD08647



Comply with EPA and Local regulations when removing or disposing of Refrigeration System oil!

START-UP PROCEDURE

Pre-Starting

Prior to starting the chiller, observe the OptiView Control Center (Refer to Form 160.54-O1). Make sure the display reads "**SYSTEM READY TO START"**.

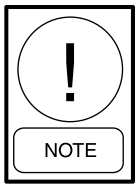


Vent any air from the chiller water boxes prior to starting the water pumps. Failure to do so will result in pass baffle damage.

START-UP

1. If the chilled water pump is manually operated, start the pump. The Control Center will not allow the chiller to start unless chilled liquid flow is established through the unit. If the chilled liquid pump is wired to the Microcomputer Control Center the pump will automatically start, therefore, this step is not necessary.
2. To start the chiller, press the **COMPRESSOR START** switch. This switch will automatically spring return to the **RUN** position. (If the unit was previously started, press the **STOP/RESET** side of the **COMPRESSOR** switch and then press the **START** side of the switch to start the chiller.) When the start switch is energized, the Control Center is placed in an operating mode and any malfunction will be noted by messages on a graphic display.

For display messages and information pertaining to the operation of the OptiView™ Control Center, refer to Form 160.54-O1.



Any malfunctions which occur during STOP/RESET are also displayed.

CHILLER OPERATION

The unit capacity will vary to maintain the leaving **CHILLED LIQUID TEMPERATURE** setpoint by the Pre-rotation Vanes which are modulated by an actuator under the control of the Microprocessor Board. The vane control routine employs proportional plus derivative (rate) control action. A drop in chilled liquid temperature will cause the actuator to close the Prerotation Vanes to decrease chiller capacity. When the chilled liquid temperature rises, the actuator will open the Prerotation Vanes to increase the capacity of the chiller.

However, the current draw (amperes) by the compressor motor is also limited to FLA setpoint by the microprocessor.

If the load continues to decrease, after the Prerotation Vanes are entirely closed, the chiller will be shut down by the Leaving Chilled Liquid – Low Temperature Control.

CONDENSER WATER TEMPERATURE CONTROL

The YORK MaxE™ chiller is designed to use less power by taking advantage of lower than design temperatures that are naturally produced by cooling towers throughout the operating year. Exact control of condenser water such as a cooling tower bypass, is not necessary for most installations. The minimum entering condenser water temperature for full and part load conditions is specified in the chiller engineering guide.

where:

$$\text{Min. ECWT} = \text{LCWT} - \text{C RANGE} + 5^{\circ}\text{F} + 12 \left(\frac{\% \text{ Load}}{100} \right)$$

$$\text{Min. ECWT} = \text{LCWT} - \text{C RANGE} + 2.8^{\circ}\text{C} + 6.6 \left(\frac{\% \text{ Load}}{100} \right)$$

ECWT = Entering Condensing Water Temperature
LCWT = Leaving Chilled Water Temperature
C Range = Condensing water temperature range at the given load condition.

At start-up, the entering condenser water temperature may be as much as 25°F (14°C) colder than the standby return chilled water temperature. Cooling tower fan cycling will normally provide adequate control of the entering condenser water temperature on most installations.

Heat recovery chillers and chillers using optional head pressure control would use a signal provided by the microprocessor to control main condenser bundle heat rejection or pressure, respectively.

OPERATING LOG SHEET

A permanent daily record of system operating conditions (temperatures and pressures) recorded at regular intervals throughout each 24 hour operating period should be kept.

An optional status printer is available for this purpose or FIG. 4 shows a log sheet used by Johnson Controls Personnel for recording test data on chiller systems. It is available from the factory in pads of 50 sheets each under Form 160.44-F7 and may be obtained through the nearest YORK office. Automatic data logging is possible by connecting the optional printer and programming the **DATA LOGGER** function.

An accurate record of readings serves as a valuable reference for operating the system. Readings taken when a system is newly installed will establish normal conditions with which to compare later readings.

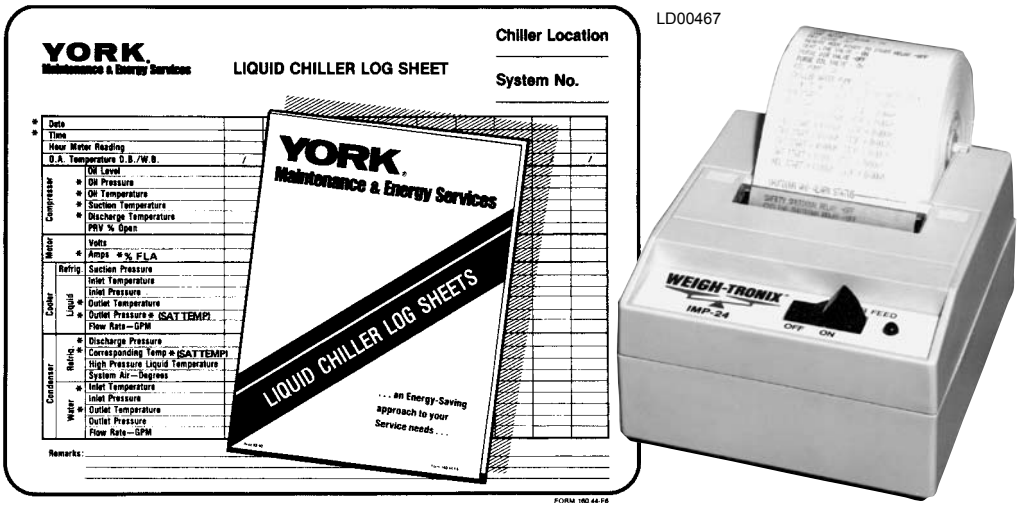
For example, an increase in condenser approach temperature (condenser temperature minus leaving condenser water temperature) may be an indication of dirty condenser tubes.

OPERATING INSPECTIONS – See Section 2

By following a regular inspection using the display readings of the Microcomputer Control Center, and maintenance procedure, the operator will avoid serious operating difficulty. The following list of inspections and procedures should be used as a guide.

Daily

1. Check OptiView™ Control Center displays.
2. If the compressor is in operation, check the bearing oil pressure on the **SYSTEM** Screen. Also check the oil level in the oil reservoir. Operating oil level should be operating range of oil indicator. Drain or add oil if necessary.
3. Check entering and leaving condenser water pressure and temperatures for comparison with job design conditions. Condenser water temperatures can be checked on the **SYSTEM** Screen.
4. Check the entering and leaving chilled liquid temperatures and evaporator pressure for comparison with job design conditions on the **SYSTEM** Screen.
5. Check the condenser saturation temperature (based upon condenser pressure sensed by the condenser transducer) on the **SYSTEM** Screen.
6. Check the compressor discharge temperature on the **SYSTEM** Screen. During normal operation discharge temperature should not exceed 220°F (104°C).
7. Check the compressor motor current on the **SYSTEM** Screen.
8. Check for any signs of dirty or fouled condenser tubes. (The temperature difference between water leaving condenser and saturated condensing temperature should not exceed the difference recorded for a new unit by more than 4°F (2.2°C)).



*__NOTE:__ These items can be printed by an electronic printer connected to the Microboard and pressing the **PRINT** key on the Keypad, or automatically using the Data Logger feature.

FIG. 4 – LIQUID CHILLER LOG SHEETS

Weekly

1. Check the refrigerant charge. (See "Checking The Refrigerant Charge", page 28.)

Monthly

1. Leak check the entire chiller.

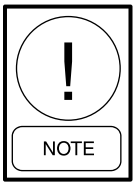
Quarterly

1. Perform chemical analysis of oil.

Semi-Annually (or more often as required)

1. Change and inspect compressor oil filter element.
2. Oil return system.
   a. Change dehydrator.
   b. Check nozzle of eductor for foreign particles.
3. Check controls and safety cutouts.

Annually (more often if necessary)



If quarterly inspection indicates oil is fine, replacing the oil is not necessary.

1. Drain and replace the oil in the compressor oil sump. (See "Oil Charging Procedure" page 21.)
2. Evaporator and Condenser.
   a. Inspect and clean water strainers.
   b. Inspect and clean tubes as required.
   c. Inspect end sheets.
3. Compressor Drive Motor (See motor manufacturers maintenance and service instruction supplied with unit)
   a. Clean air passages and windings per manufacturers instructions.
   b. Meg motor windings – See FIG. 11 for details.
   c. Lubricate per motor manufacturer recommendations.
4. Inspect and service electrical components as necessary.
5. Perform refrigerant analysis.

NEED FOR MAINTENANCE OR SERVICE

If the system is malfunctioning in any manner or the unit is stopped by one of the safety controls, consult the "Operation Analysis Chart", (Table 1), pages 22 and 23 of this instruction. After consulting this chart, if you are unable to make the proper repairs or adjustments to start the compressor or the particular trouble continues to hinder the performance of the unit, please call the nearest Johnson Controls District Office. Failure to report constant troubles could damage the unit and increase the cost of repairs.

STOPPING THE SYSTEM

The Optiview™ Control Center can be programmed to start and stop automatically (maximum, once each day) whenever desired. Refer to Form 160.54-O1. To stop the chiller, proceed as follows:

1. Push the soft shutdown key on the homescreen on the OptiView panel or rapid stop with the **COMPRESSOR STOP/RESET** switch. The compressor will stop automatically. The oil pump will continue to run for coastdown period. The oil pump will then stop automatically.
2. Stop the chilled water pump (if not wired into the Microcomputer Control Center, in which case it will shut off automatically simultaneously with the oil pump.) (The actual water pump contact operation is dependent upon the position of Microboard jumper J54.)
3. Open the switch to the cooling tower fan motors, if used.
4. The compressor sump oil heater is energized when the unit is stopped.

PROLONGED SHUTDOWN

If the chiller is to be shut down for an extended period of time (for example, over the winter season), the following paragraphs outline the procedure to be followed.

1. Test all system joints for refrigerant leaks with a leak detector. If any leaks are found, they should be repaired before allowing the system to stand for a long period of time.

   During long idle periods, the tightness of the system should be checked periodically.

2. If freezing temperatures are encountered while the system is idle, carefully drain the cooling water from the cooling tower, condenser, condenser pump, and the chilled water system-chilled water pump and coils.

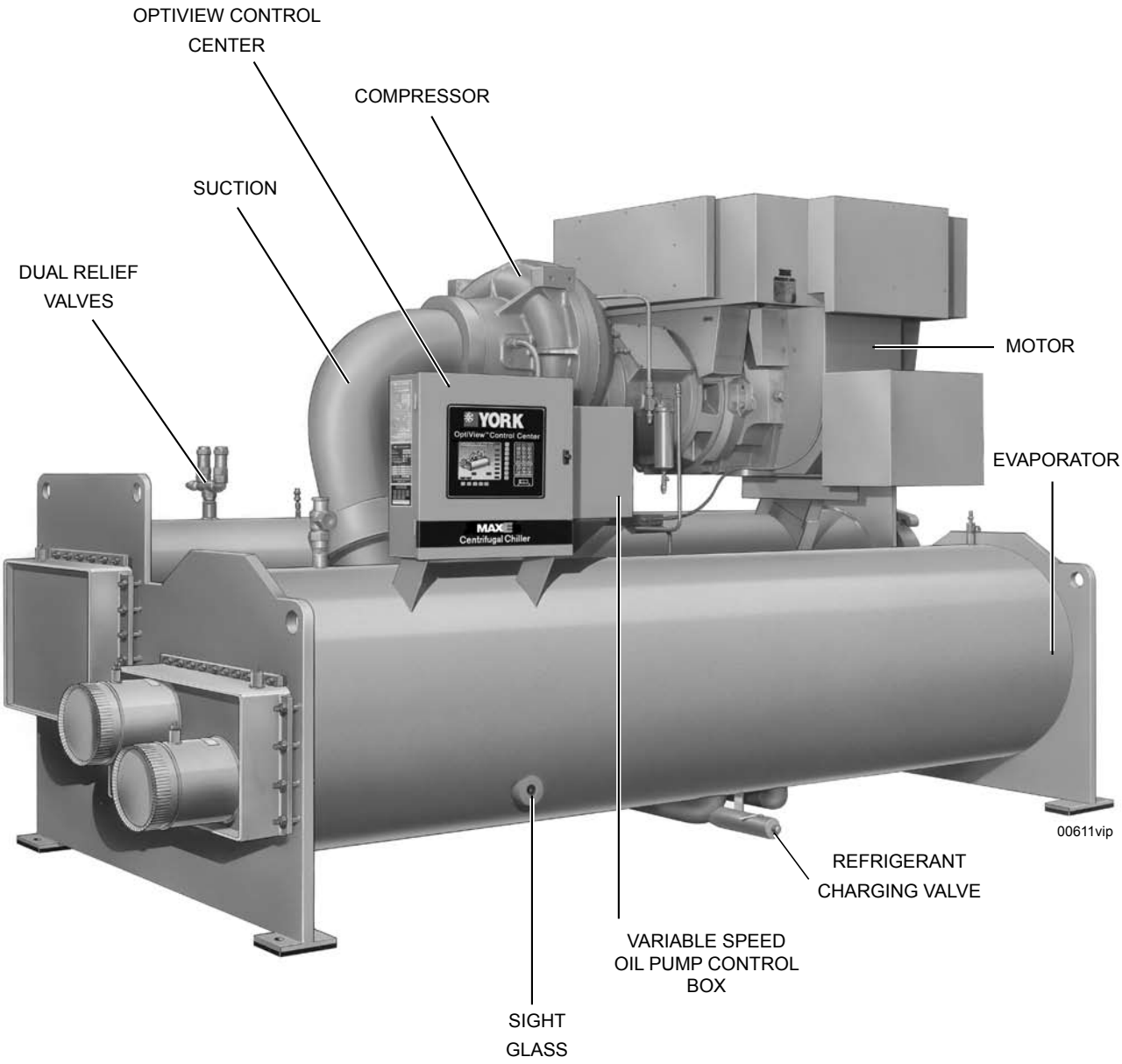
   Open the drains on the evaporator and condenser liquid heads to assure complete drainage. (If a Variable Speed Drive, drain its cooling system. If Solid State Starter equipped drain liquid from starter cooling loop.)

3. If freezing temperatures are encountered for periods longer than a few days, the refrigerant should be recovered to containers to prevent leakage from O-ring joints.

4. On the **SETUP** Screen, disable the clock. This conserves the battery.

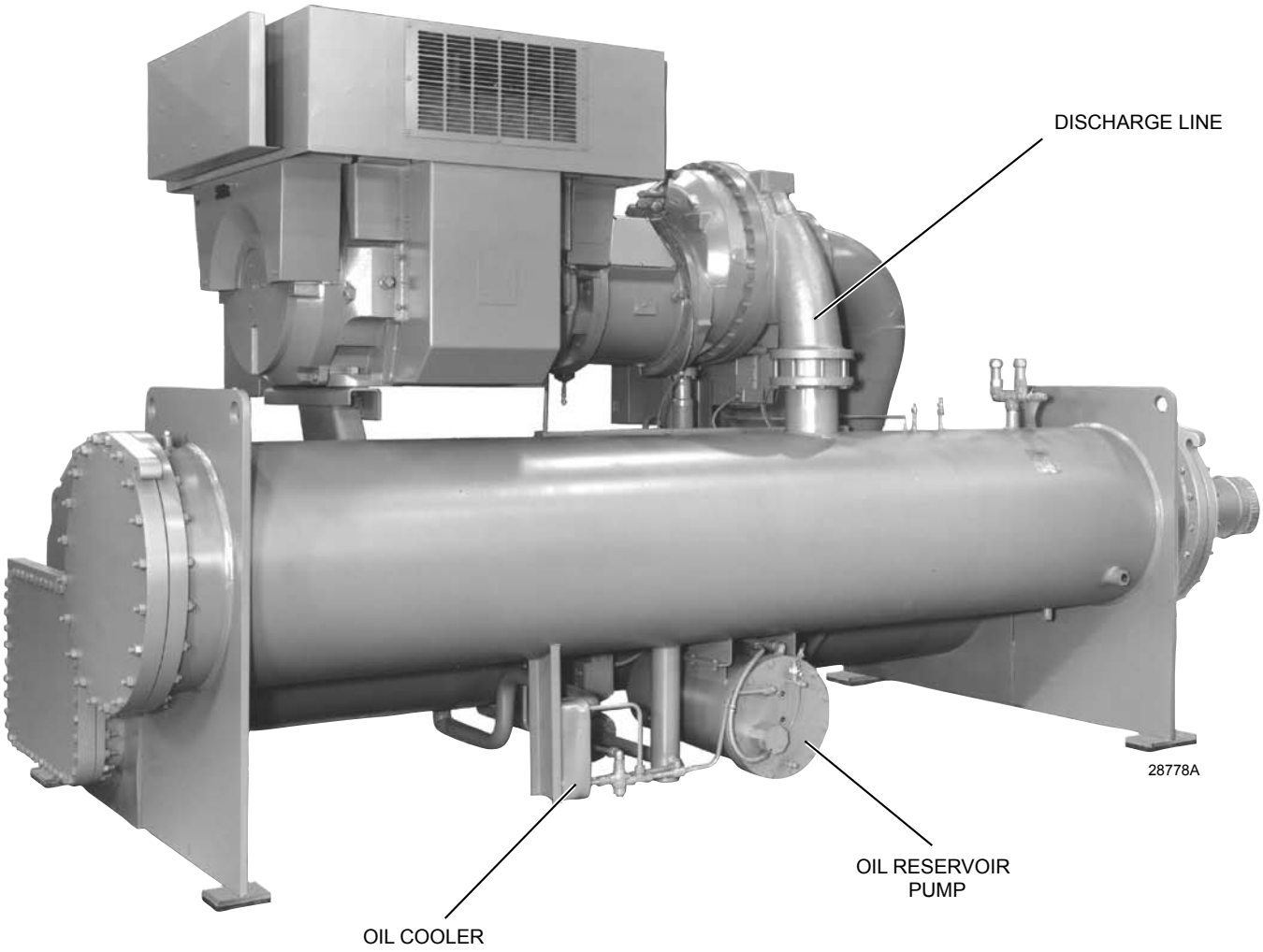
5. Open the main disconnect switches to the compressor motor, condenser water pump and the chilled water pump. Open the 115 volt circuit to the Control Center.

SECTION 3
SYSTEM COMPONENTS DESCRIPTION



FRONT VIEW

FIG. 5 – SYSTEM COMPONENTS



REAR VIEW

FIG. 5 – SYSTEM COMPONENTS (CONT'D)

GENERAL

The YORK Model YK MaxE™ Centrifugal Liquid Chiller is completely factory-packaged including evaporator, condenser, compressor, motor, lubrication system, OptiView Control Center, and all interconnecting unit piping and wiring.

COMPRESSOR

The compressor is a single-stage centrifugal type powered by an open-drive electric motor.

The rotor assembly consists of a heat-treated alloy steel drive shaft and impeller shaft with a cast aluminum, fully shrouded impeller. The impeller is designed for balanced thrust and is dynamically balanced and over-speed tested. The inserted type journal and thrust bearings are fabricated of aluminum alloy. Single helical gears with crowned teeth are designed so that more than one tooth is in contact at all times. Gears are integrally assembled in the compressor rotor support and are film lubricated. Each gear is individually mounted in its own journal and thrust bearings.

The open-drive compressor shaft seal is a double bellows cartridge style with ceramic internal and atmospheric seal faces. The seal is oil-flooded at all times and is pressure-lubricated during operation.

CAPACITY CONTROL

Prerotation vanes (PRV) modulate chiller capacity from 100% to as low as 15% of design for normal air conditioning applications. Operation is by an external, electric PRV actuator which automatically controls the vane position to maintain a constant leaving chilled liquid temperature.

COMPRESSOR LUBRICATION SYSTEM (See FIG. 6)

The chiller lubrication system consists of the oil pump, oil filter, oil cooler and all interconnecting oil piping and passages. There are main points within the compressor which must be supplied with forced lubrication as follows:

1. **Compressor Drive Shaft (Low Speed)**
   a. Shaft seal.
   b. Front and rear journal bearings – one on each side of driving gear.
   c. Low speed thrust bearing (forward and reverse).
2. **Compressor Driven Shaft (High Speed)**
   a. Forward and reverse high speed thrust bearing.
   b. Two journal bearings.
3. **Speed Increasing Gears**
   a. Meshing surfaces of drive and pinion gear teeth.

To provide the required amount of oil under the necessary pressure to properly lubricate these parts, a motor driven submersible oil pump is located in a remote oil sump.

Upon pressing of the **COMPRESSOR START** switch on the Control Center, the oil pump is immediately energized. After a 50 second pre-lube period, the compressor motor will start. The oil pump will continue to run during the entire operation of the compressor, and for 150 seconds during compressor coastdown.

The submerged oil pump takes suction from the surrounding oil and discharges it to the oil cooler where heat is rejected. The oil flows from the oil cooler to the oil filter. The oil leaves the filter and flows to the emergency oil reservoir where it is distributed to the compressor bearings. The oil lubricates the compressor rotating components and is returned to the oil sump.

There is an emergency oil reservoir located at the highest point in the lubrication system internally in the compressor. It provides an oil supply to the various bearings and gears in the event of a system shutdown due to power failure. The reservoir, located on the top of the compressor, allows the oil to be distributed through the passages by gravity flow, thus providing necessary lubrication during the compressor coastdown.

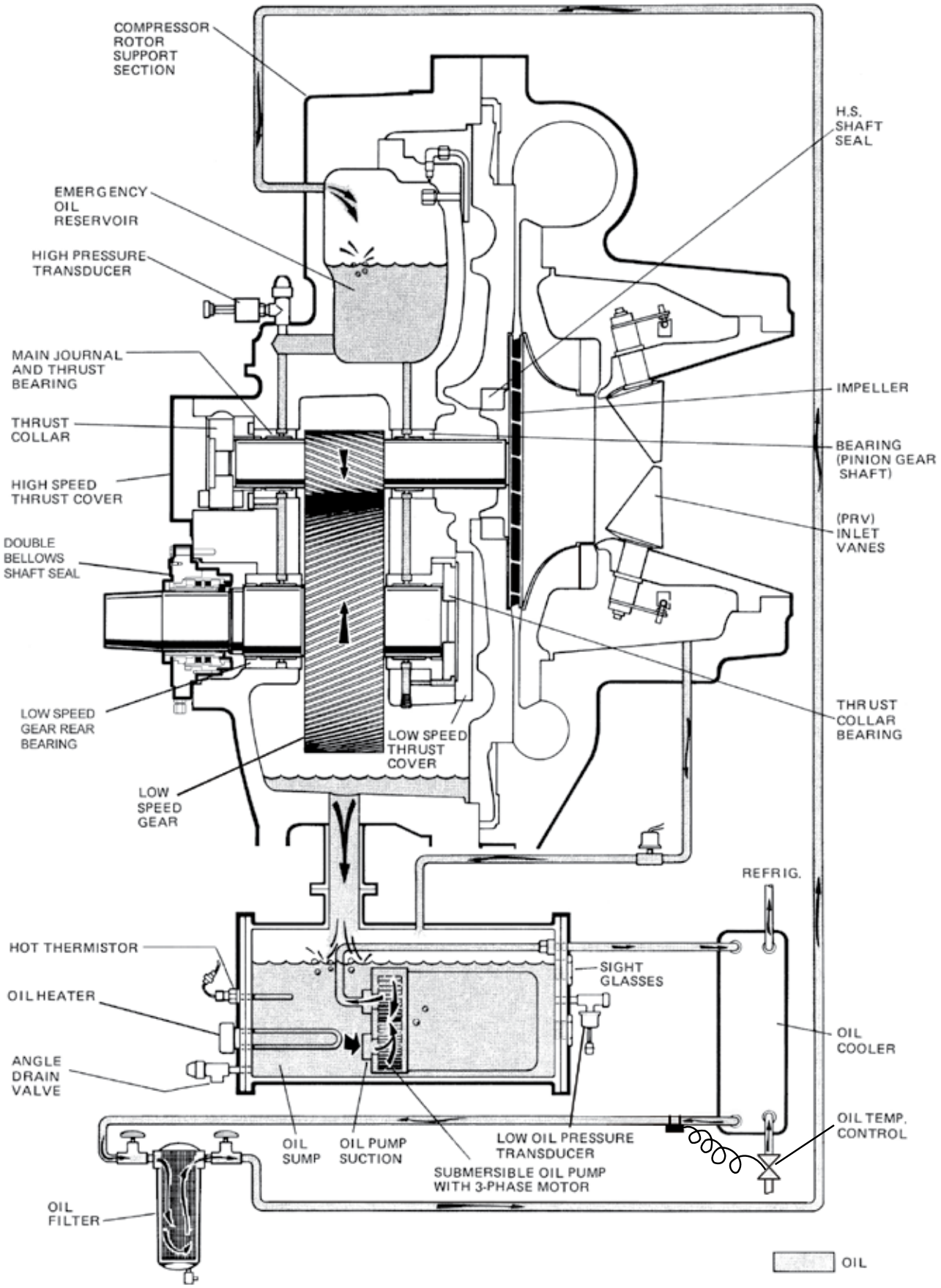


FIG. 6 – SCHEMATIC DRAWING – (YK) COMPRESSOR LUBRICATION SYSTEM

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OIL PUMP

For normal operation, the oil pump should operate at all times during chiller operation.

On shutdown of the system for any reason, the oil pump operates and continues to run for 150 seconds. The system cannot restart during that time interval.

OIL HEATER

During long idle periods, the oil in the compressor oil reservoir tends to absorb as much refrigerant as it can hold, depending upon the temperature of the oil and the pressure in the reservoir. As the oil temperature is lowered, the amount of refrigerant absorbed will be increased. If the quantity of refrigerant in the oil becomes excessive, violent oil foaming will result as the pressure within the system is lowered on starting. This foaming is caused by refrigerant boiling out of the oil as the pressure is lowered. If this foam reaches the oil pump suction, the bearing oil pressure will fluctuate with possible temporary loss of lubrication, causing the oil pressure safety cutout to actuate and stop the system. See “Control Center” Form 160.54-O1.

MOTOR DRIVELINE

The compressor motor is an open-drip-proof, squirrel cage, induction type constructed to YORK design specifications. 60 hertz motors operate at 3570 rpm. 50 hertz motors operate at 2975 rpm.

The open motor is provided with a D-flange, cast iron adapter mounted to the compressor and supported by a motor support.

Motor drive shaft is directly connected to the compressor shaft with a flexible disc coupling. This coupling has all metal construction with no wearing parts to assure long life, and no lubrication requirements to provide low maintenance.

For units utilizing remote Electro-Mechanical starters, a terminal box is provided for field connected conduit. Motor terminals are brought through the motor casing into the terminal box. Jumpers are furnished for three-lead type of starting. Motor terminal lugs are not furnished. Overload/overcurrent transformers are furnished with all units.

HEAT EXCHANGERS

Evaporator and condenser shells are fabricated from rolled carbon steel plates with fusion welded seams.

Heat exchanger tubes are internally enhanced type.

The evaporator is a shell and tube, flooded type heat exchanger. A distributor trough provides uniform distribution of refrigerant over the entire shell length. Stainless steel mesh eliminators or suction baffles are located above the tube bundle to prevent liquid refrigerant carryover into the compressor. A 2" liquid level sight glass is located on the side of the shell to aid in determining proper refrigerant charge. The evaporator shell contains dual refrigerant relief valves unless condenser isolation is installed.

The condenser is a shell and tube type, with a discharge gas baffle to prevent direct high velocity impingement on the tubes. A separate subcooler (except heat recovery chillers) is located in the condenser to enhance performance. Dual refrigerant relief valves are located on condenser shells and optional refrigerant isolation valves are available.

The removable compact water boxes are fabricated of steel. The design working pressure is 150 PSIG (1034 kPa) and the boxes are tested at 225 PSIG (1551 kPa). Integral steel water baffles provide the required pass arrangements. Stub-out water nozzle connections with Victaulic grooves are welded to the water boxes. These nozzle connections are suitable for Victaulic couplings, welding or flanges, and are capped for shipment. Plugged 3/4" drain and vent connections are provided in each water box. Optional marine waterboxes are available.

REFRIGERANT FLOW CONTROL

Refrigerant flow to the evaporator is controlled by a variable orifice.

A level sensor senses the refrigerant level in the condenser and outputs an analog voltage to the Microboard that represents this level (0% = empty; 100% = full). Under program control, the Microboard modulates a variable orifice to control the condenser refrigerant level to a programmed setpoint. Other setpoints affect the control sensitivity and response. These setpoints must be entered at chiller commissioning by a qualified service technician. Only a qualified service technician may modify these settings.

While the chiller is shut down, the orifice will be in the fully open position causing the sensed level to be approximately 0%. When the chiller is started, after the vane motor end switch (VMS) opens when entering **SYSTEM RUN**, if actual level is less than the level setpoint, a linearly increasing ramp is applied to the level setpoint. This ramp causes the setpoint to go from the initial refrigerant level (approximately 0%) to the programmed setpoint over a programmable period of time.

If the actual level is greater than the setpoint when the VMS opens, there is no pulldown period, it immediately begins to control to the programmed setpoint.

While the chiller is running, the refrigerant level is normally controlled to the level setpoint. However, anytime the vanes fully close (VMS closes), normal level control is terminated, any refrigerant level setpoint in effect is cancelled and the outputs to the level control will be constant open.

OPTIONAL SERVICE ISOLATION VALVES

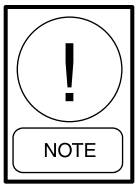
If your chiller is equipped with optional service isolation valves on the discharge and liquid line, these valves must remain open during operation. These valves are used for isolating the refrigerant charge in either the evaporator or condenser to allow service access to the system. A refrigerant pump-out unit will be required to isolate the refrigerant.



Isolation of the refrigerant in this system must be performed by a qualified service technician.

OPTIONAL HOT GAS BYPASS

Hot gas bypass is optional and is used to eliminate compressor surge during light load or high head operation. The OptiView control panel will automatically modulate the hot gas valve open and closed as required. Adjustment of the hot gas control valve must be performed by a qualified service technician following the Hot Gas Set-up procedure.



Changes in chilled water flow will require re-adjustment of the hot gas control to insure proper operation.

OTIVIEW CONTROL CENTER (See Section 2)

The OptiView Control Center is factory-mounted, wired and tested. The electronic panel automatically controls the operation of the unit in meeting system cooling requirements while minimizing energy usage. For detailed information on the Control Center, refer to “Section 2” of this manual.

SOLID STATE STARTER (Optional)

The Solid State Starter is a reduced voltage starter that controls and maintains a constant current flow to the motor during start-up. It is mounted on the chiller. Power and control wiring between the starter and chiller are factory installed. The starter enclosure is NEMA-1 with a hinged access door with lock and key. Electrical lugs for incoming power wiring are provided.

VARIABLE SPEED DRIVE (Optional)

A Variable Speed Drive can be factory packaged with the chiller. It is designed to vary the compressor motor speed and prerotation vane position by controlling the frequency and voltage of the electrical power to the motor. Operational information is contained in Form 160.00-O1. The control logic automatically adjusts motor speed and compressor prerotation vane position for maximum part load efficiency by analyzing information fed to it by sensors located throughout the chiller.

SECTION 4
OPERATIONAL MAINTENANCE

OIL RETURN SYSTEM

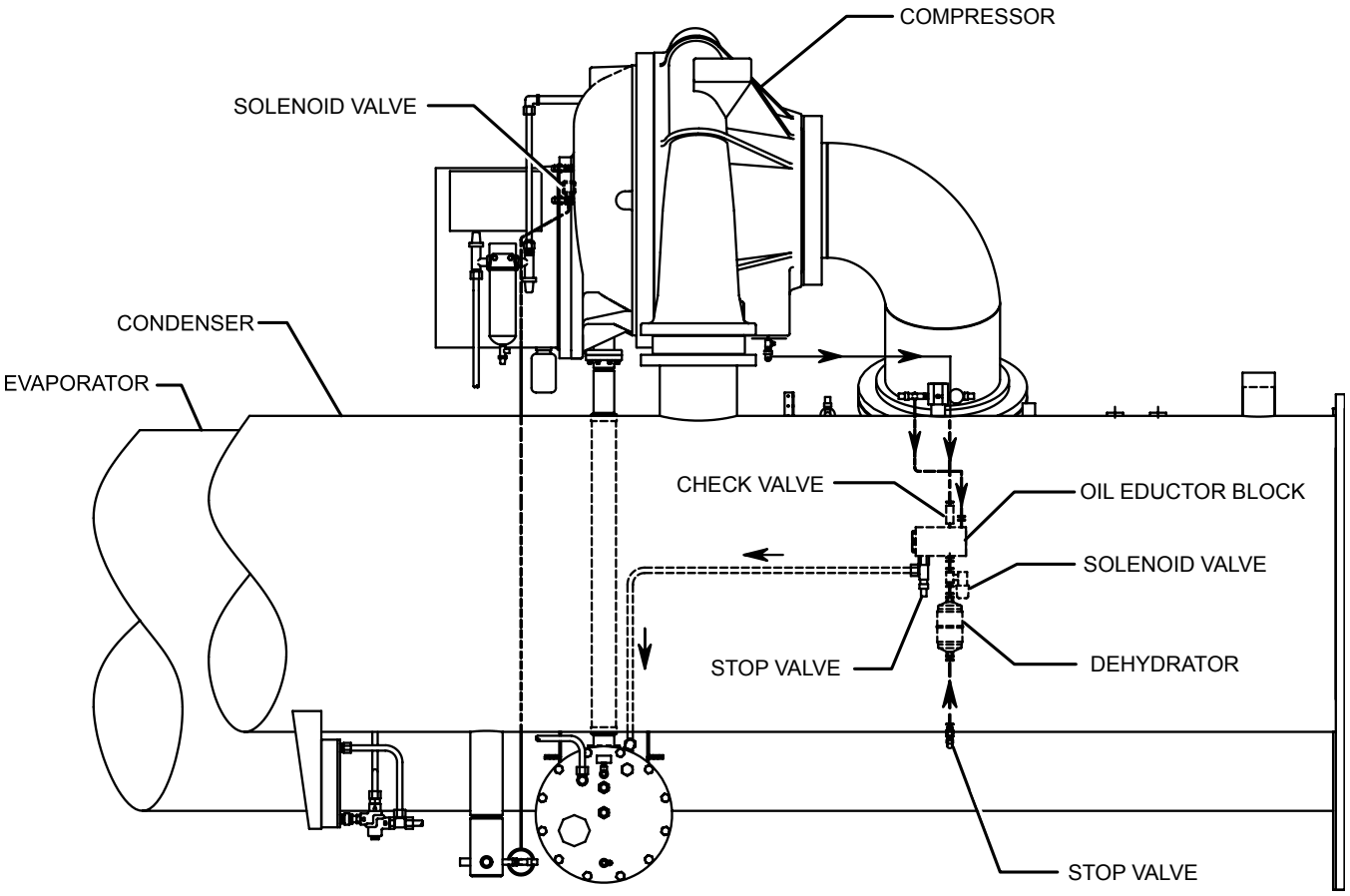
The oil return system continuously maintains the proper oil level in the compressor oil sump (See FIG. 7).

High pressure condenser gas flows continuously through the eductor inducing the low pressure, oil rich liquid to flow from the evaporator, through the dehydrator to the compressor sump.

CHANGING THE DEHYDRATOR

To change the dehydrator, use the following procedure:

1. Isolate the dehydrator at the stop valves.
2. Remove the dehydrator (See to FIG. 7).
3. Assemble the new filter-drier.
4. Open evaporator stop valve and check dehydrator connections for refrigerant leaks.
5. Open all the dehydrator stop valves.



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FIG. 7 – OIL RETURN SYSTEM

THE OIL CHARGE

The nominal oil charge for all H, K, & P8-P9 YK compressors is 20 gallons, and for Q3-Q7 YK compressors is 10 gallons of type “YORK K”.

New YORK Refrigeration oil must be used in the centrifugal compressor. Since oil absorbs moisture when exposed to the atmosphere, it should be kept tightly capped until used.

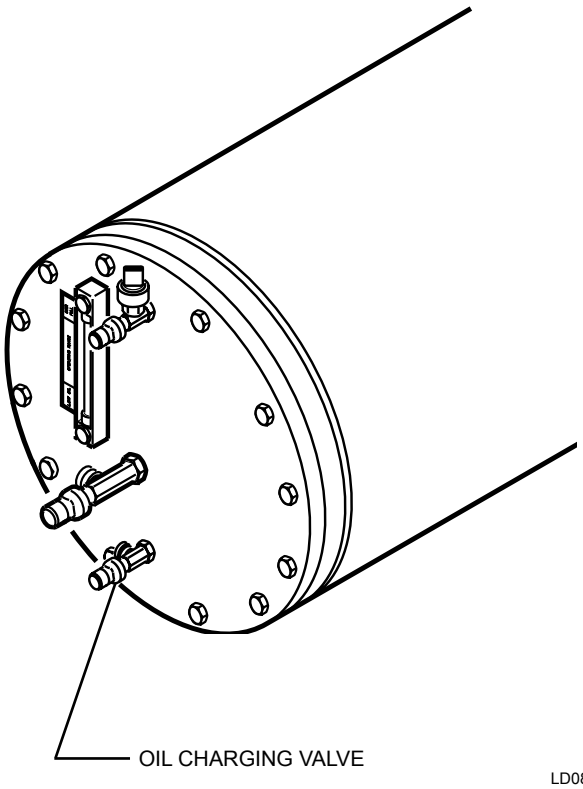
OIL CHARGING PROCEDURE

During operation the compressor oil level must be maintained in the "Operating Range" identified on the vertical oil level indicator. If the oil level falls into the lower sight glass, it is necessary to add oil to the compressor oil reservoir. The oil should be charged into the oil reservoir using the YORK Oil Charging Pump – YORK Part No. 070-10654. To charge oil into the oil reservoir, proceed as follows:

1. The unit must be shut down.
2. Immerse the suction connection of the oil charging pump in a clean container of new oil and connect the pump discharge connection to the oil charging valve (A) located on the remote oil reservoir cover plate (See FIG. 8.) Do not tighten the connection at the charging valve until after the air is forced out by pumping a few strokes of the oil pump. This fills the lines with oil and prevents air from being pumped into the system.
3. Open the oil charging valve and pump oil into the system until oil level in the compressor oil reservoir is in the “Over Full” region of the oil level indicator label. Close the charging valve and disconnect the hand oil pump.
4. As soon as oil charging is complete, close the power supply to the starter to energize the oil heater. This will keep the concentration of refrigerant in the oil to a minimum.

When the oil reservoir is initially charged with oil, the oil pump should be started manually to fill the lines, passages, oil cooler and oil filter. This will lower the oil level in the reservoir. It may then be necessary to add oil to bring the level back into the “Operating Range” of the oil level indicator label.

4



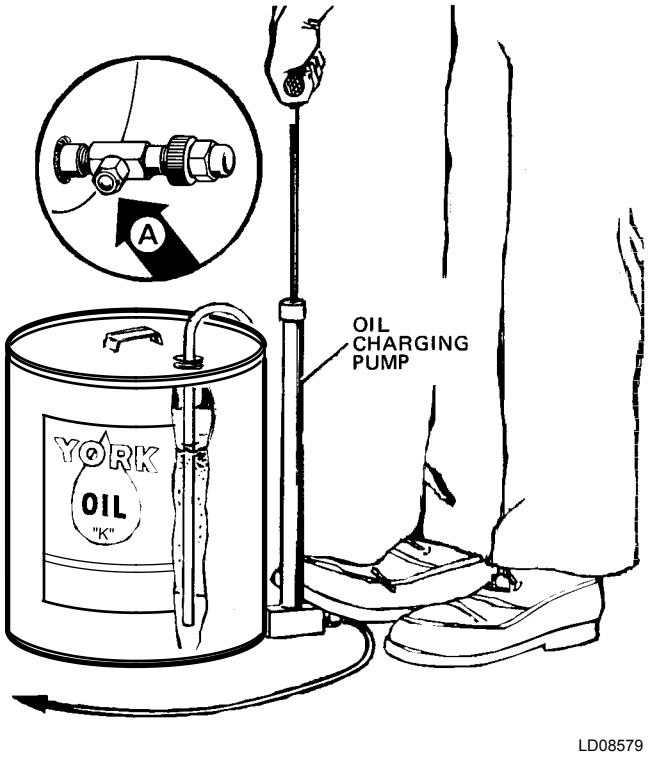


FIG. 8 – CHARGING OIL RESERVOIR WITH OIL

SECTION 5 – TROUBLESHOOTING

TABLE 1 – OPERATION ANALYSIS CHART

RESULTS	POSSIBLE CAUSE	REMEDY
1. SYMPTOM: ABNORMALLY HIGH DISCHARGE PRESSURE		
Temperature difference between condensing temperature and water off condenser higher than normal.	Air in condenser.	
High discharge pressure.	Condenser tubes dirty or scaled	Clean condenser tubes. Check water conditioning.
	High condenser water temperature.	Reduce condenser water inlet temperature. (Check cooling tower and water circulation.)
Temperature difference between condenser water on and water off higher than normal, with normal evaporator pressure.	Insufficient condensing water flow.	Increase the quantity of water through the condenser to proper value.
2. SYMPTOM: ABNORMALLY LOW SUCTION PRESSURE		
Temperature difference between leaving chilled water and refrigerant in evaporator greater than normal with high discharge temperature.	Insufficient charge of refrigerant.	Check for leaks and charge refrigerant into system.
	Variable orifice problem.	Remove obstruction·
Temperature difference between leaving chilled water and refrigerant in the evaporator greater than normal with normal discharge temperature.	Evaporator tubes dirty or restricted.	Clean evaporator tubes.
Temperature of chilled water too low with with low motor amperes.	Insufficient load for system capacity.	Check prerotation vane motor operation and setting of low water temperature cutout.
3. SYMPTOM: HIGH EVAPORATOR PRESSURE		
High chilled water temperature.	Prerotation vanes fail to open.	Check the prerotation vane motor positioning circuit.
	System overload.	Be sure the vanes are wide open (without overloading the motor) until the load decreases.
4. SYMPTOM: NO OIL PRESSURE WHEN SYSTEM START BUTTON PUSHED		
Low oil pressure displayed on control center; compressor will not start.	Oil pump running in wrong direction.	Check rotation of oil pump (Electrical Connections).
	Oil pump not running.	Troubleshoot electrical problem with oil pump VSD.

TABLE 1 – OPERATION ANALYSIS CHART (CONTINUED)

RESULTS	POSSIBLE CAUSE	REMEDY
5. SYMPTOM: UNUSUALLY HIGH OIL PRESSURE DEVELOPS WHEN OIL PUMP RUNS		
Unusually high oil pressure is displayed when the oil pressure display key is pressed when the oil pump is running.	High oil pressure. Transducer defective.	Replace low or high oil pressure transducer.
6. SYMPTOM: OIL PUMP VIBRATES OR IS NOISY		
Oil pump vibrates or is extremely noisy with some oil pressure when pressing **OIL PRESSURE** display key.	Oil not reaching pump suction inlet in sufficient quantity.	Check oil level.
NOTE: ***When oil pump is run without an oil supply it will vibrate and become extremely noisy.***	Worn or failed oil pump.	Repair/Replace oil pump.
7. SYMPTOM: REDUCED OIL PUMP CAPACITY		
Oil pump pumping capacity.	Excessive end clearance pump. Other worn pump parts.	Inspect and replace worn parts.
	Partially blocked oil supply inlet.	Check oil inlet for blockage.
8. SYMPTOM: OIL PRESSURE GRADUALLY DECREASES (Noted by Observation of Daily Log Sheets)		
When oil pump VSD frequency increases to 55 + hz to maintain target oil pressure.	Oil filter is dirty.	Change oil filter.
9. SYMPTOM: OIL PRESSURE SYSTEM CEASES TO RETURN AN OIL/REFRIGERANT SAMPLE		
Oil refrigerant return not functioning.	Filter-drier in oil return system dirty.	Replace old filter-drier with new.
	Jet or orifice of oil return jet clogged.	Remove jet, inspect for dirt. Remove dirt using solvent and replace.
10. SYMPTOM: OIL PUMP FAILS TO DELIVER OIL PRESSURE		
No oil pressure registers when pressing **OIL PRESSURE** display key when oil pump runs.	Faulty oil pressure transducer Faulty wiring/connectors.	Replace oil pressure transducer.

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SECTION 6
MAINTENANCE

RENEWAL PARTS

For any required Renewal Parts, refer to YORK Renewal Parts Unit Components Manual 160.75-RP1.

CHECKING SYSTEM FOR LEAKS

Leak Testing During Operation
The refrigerant side of the system is carefully pressure tested and evacuated at the factory.

After the system has been charged, the system should be carefully leak tested with a R-134a compatible leak detector to be sure all joints are tight.

If any leaks are indicated, they must be repaired immediately. Usually, leaks can be stopped by tightening flare nuts or flange bolts. However, for any major repair, the refrigerant charge must be removed. (See "Handling Refrigerant for Dismantling and Repair", page 29.)

CONDUCTING R-22 PRESSURE TEST

With the R-134a charge removed and all known leaks repaired, the system should be charged with a small amount of R-22 mixed with dry nitrogen so that a halide torch or electronic leak detector can be used to detect any leaks too small to be found by the soap test.

To test with R-22, proceed as follows:

1. With no pressure in the system, charge R-22 gas into the system through the charging valve to a pressure of 2 PSIG (14 kPa).

2. Build up the system pressure with dry nitrogen to approximately 75 to 100 PSIG (517 to 690 kPa). To be sure that the concentration of refrigerant has reached all part of the system, slightly open the oil charging valve and test for the presence of refrigerant with a leak detector.

3. Test around each joint and factory weld. It is important that this test be thoroughly and carefully done, spending as much time as necessary and using a good leak detector.

4. To check for refrigerant leaks in the evaporator and condenser, open the vents in the evaporator and condenser heads and test for the presence of refrigerant. If no refrigerant is present, the tubes and tube sheets may be considered tight. If refrigerant is detected at the vents, the heads must be removed, the leak located (by means of soap test or leak detector) and repaired.

EVACUATION AND DEHYDRATION OF UNIT

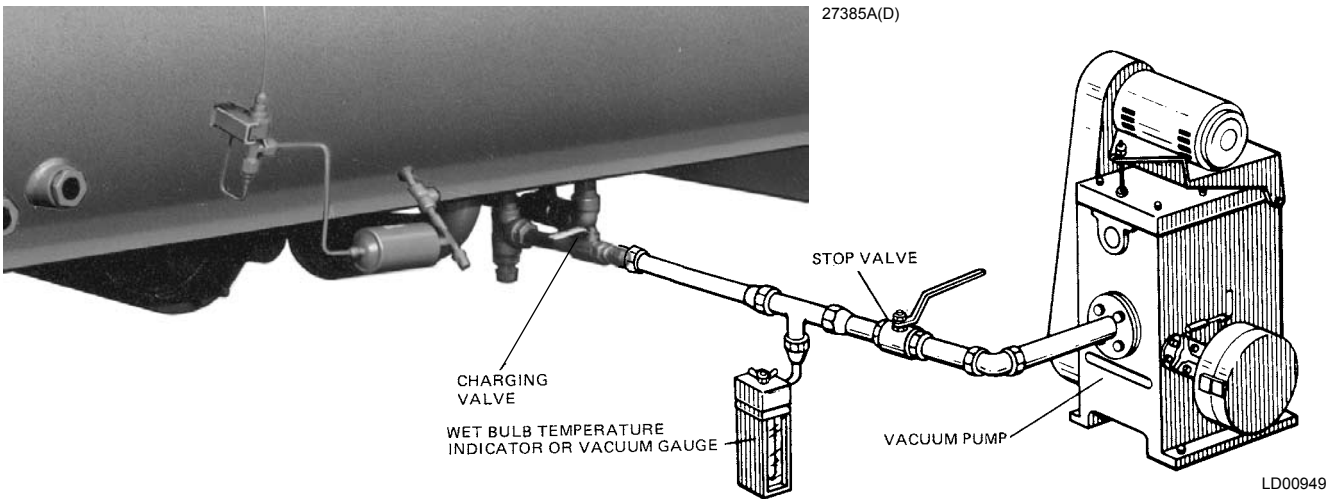


FIG. 9 – EVACUATION OF CHILLER

*GAUGE	ABSOLUTE			BOILING TEMPERATURES OF WATER °F
INCHES OF MERCURY (HG) BELOW ONE STANDARD ATMOSPHERE	PSIA	MILLIMETERS OF MERCURY (HG)	MICRONS	
0	14.696	760.	760,000	212
10.24"	9.629	500.	500,000	192
22.05"	3.865	200.	200,000	151
25.98"	1.935	100.	100,000	124
27.95"	.968	50.	50,000	101
28.94"	.481	25.	25,000	78
29.53"	.192	10.	10,000	52
29.67"	.122	6.3	6,300	40
29.72"	.099	5.	5,000	35 ——— WATER FREEZES
29.842"	.039	2.	2,000	15
29.882"	.019	1.0	1,000	+1
29.901"	.010	.5	500	–11
29.917"	.002	.1	100	–38
29.919"	.001	.05	50	–50
29.9206"	.0002	.01	10	–70
29.921"	0	0	0	

*One standard atmosphere = 14.696 PSIA
= 760 mm Hg. absolute pressure at 32°F
= 29.921 inches Hg. absolute at 32°F

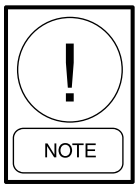
NOTES: PSIA = Lbs. per sq. in. gauge pressure
= Pressure above atmosphere
PSIA = Lbs. per sq. in. absolute pressure
= Sum of gauge plus atmospheric pressure

TABLE 2 – SYSTEM PRESSURES

VACUUM TESTING

After the pressure test has been completed, the vacuum test should be conducted as follows:

1. Connect a high capacity vacuum pump, with indicator, to the system charging valve as shown in FIG. 9 and start the pump. (See "Vacuum Dehydration".)
2. Open wide all system valves. Be sure all valves to the atmosphere are closed.
3. Operate the vacuum pump in accordance with **VACUUM DEHYDRATION** until a wet bulb temperature of +32°F or a pressure of 5 mm Hg is reached. See Table 2 for corresponding values of pressure.
4. To improve evacuation circulate hot water (not to exceed 125°F, 51.7ºC) through the evaporator and condenser tubes to thoroughly dehydrate the shells. If a source of hot water is not readily available, a portable water heater should be employed. **DO NOT USE STEAM**. A suggested method is to connect a hose between the source of hot water under pressure and the evaporator head drain connection, out the evaporator vent connection, into the condenser head drain and out the condenser vent. To avoid the possibility of causing leaks, the temperature should be brought up slowly so that the tubes and shell are heated evenly.
5. Close the system charging valve and the stop valve between the vacuum indicator and the vacuum pump. Then disconnect the vacuum pump leaving the vacuum indicator in place.
6. Hold the vacuum obtained in Step 3 in the system for 8 hours; the slightest rise in pressure indicates a leak or the presence of moisture, or both. If, after 24 hours the wet bulb temperature in the vacuum indicator has not risen above 40°F (4.4°C) or a pressure of 6.3 mm Hg, the system may be considered tight.



Be sure the vacuum indicator is valved off while holding the system vacuum and be sure to open the valve between the vacuum indicator and the system when checking the vacuum after the 8 hour period.

7. If the vacuum does not hold for 8 hours within the limits specified in Step 6 above, the leak must be found and repaired.

VACUUM DEHYDRATION

To obtain a sufficiently dry system, the following instructions have been assembled to provide an effective method for evacuating and dehydrating a system in the field. Although there are several methods of dehydrating a system, we are recommending the following, as it produces one of the best results, and affords a means of obtaining accurate readings as to the extent of dehydration.

The equipment required to follow this method of dehydration consists of a wet bulb indicator or vacuum gauge, a chart showing the relation between dew point temperature and pressure in inches of mercury (vacuum), (See Table 2) and a vacuum pump capable of pumping a suitable vacuum on the system.

OPERATION

Dehydration of a refrigerant system can be obtained by this method because the water present in the system reacts much as a refrigerant would. By pulling down the pressure in the system to a point where its saturation temperature is considerably below that of room temperature, heat will flow from the room through the walls of the system and vaporize the water, allowing a large percentage of it to be removed by the vacuum pump. The length of time necessary for the dehydration of a system is dependent on the size or volume of the system, the capacity and efficiency of the vacuum pump, the room temperature and the quantity of water present in the system. By the use of the vacuum indicator as suggested, the test tube will be evacuated to the same pressure as the system, and the distilled water will be maintained at the same saturation temperature as any free water in the system, and this temperature can be observed on the thermometer.

If the system has been pressure tested and found to be tight prior to evacuation, then the saturation temperature recordings should follow a curve similar to the typical saturation curve shown as FIG. 10.

The temperature of the water in the test tube will drop as the pressure decreases, until the boiling point is reached, at which point the temperature will level off and remain at this level until all of the water in the shell is vaporized. When this final vaporization has taken place the pressure and temperature will continue to drop until eventually a temperature of 35°F (1.6°C) or a pressure of 5 mm Hg. is reached.

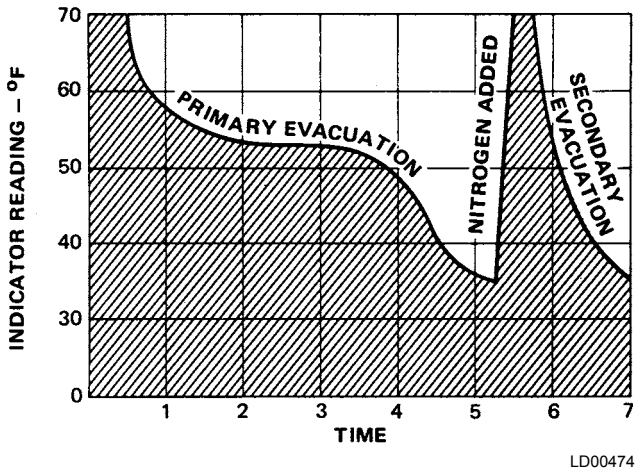


FIG. 10 – SATURATION CURVE

When this point is reached, practically all of the air has been evacuated from the system, but there is still a small amount of moisture left. In order to provide a medium for carrying this residual moisture to the vacuum pump, nitrogen should be introduced into the system to bring it to atmospheric pressure and the indicator temperature will return to approximately ambient temperature. Close off the system again, and start the second evacuation.

The relatively small amount of moisture left will be carried out through the vacuum pump and the temperature or pressure shown by the indicator should drop uniformly until it reaches a temperature of 35°F (1.6°C) or a pressure of 5 mm Hg.

When the vacuum indicator registers this temperature or pressure, it is a positive sign that the system is evacuated and dehydrated to the recommended limit. If this level cannot be reached, it is evident that there is a leak somewhere in the system. Any leaks must be corrected before the indicator can be pulled down to 35°F or 5 mm Hg. in the primary evacuation.

During the primary pulldown, keep a careful watch on the wet bulb indicator temperature, and do not let it fall below 35°F (1.6°C). If the temperature is allowed to fall to 32°F (0°C), the water in the test tube will freeze, and the result will be a faulty temperature reading.

REFRIGERANT CHARGING

To avoid the possibility of freezing liquid within the evaporator tubes when charging an evacuated system, only refrigerant vapor from the top of the drum or cylinder must be admitted to the system pressure until the system pressure is raised above the point corresponding to the freezing point of the evaporator liquid. For water, the pressure corresponding to the freezing point is 8.54 PSIG (58.9 kPa) for R-134a (at sea level).

While charging, every precaution must be taken to prevent moisture laden air from entering the system. Make up a suitable charging connection from new copper tubing to fit between the system charging valve and the fitting on the charging drum. This connection should be as short as possible but long enough to permit sufficient flexibility for changing drums. The charging connection should be purged each time a full container of refrigerant is connected and changing containers should be done as quickly as possible to minimize the loss of refrigerant.

Refrigerant may be furnished in cylinders containing either 30, 50, 125, 1,025 or 1750 lbs. (13.6, 22.6, 56.6, 464 or 794 kg) of refrigerant.

TABLE 3 – REFRIGERANT CHARGE

Evap Code	Cond Code	Factory Charged Lbs.
AP	AP, AQ, AR, AS	828
AQ		794
AR		739
AS		752
CP	BW, BX	1525
CQ		1460
CR		1400
CS		1400
CP	CP, CQ, CR, CS	1221
CQ		1185
CR		1100
CS		1091
DP	DP, DQ, DR, DS	1628
DQ		1580
DR		1466
DS		1455
EP	EP, EQ, ER, ES, ET	1721
EQ		1718
ER		1635
ES		1574
ET		1474
EP	IW, IX	1805
EQ		1775
ER		1710
ES		1680
ET		1615
FQ	FQ, FR, FS, FT	2162
FR		2160
FS		2099
FT		1966
GQ	EV, EW, EX	1990
GR		1899
GS		1808
HQ	FV, FW, FX	2626
HR		2506
HS		2385
JP	JP, JQ, JR, JS	2557
JQ		2401
JR		2143
JS		2051
LQ	LQ, LR, LS	3175
LR		2857
LS		2734
KP	KP, KQ, KR, KS K2, K3, K4	2920
KQ		2742
KR		2490
KS		2384
KT		2992
KV		2808
KW		2571
KX		2462
K2		3146
K3		3065
K4		2983
K5		3248
K6		3165
K7		3081
MQ	UW, UX U8, U9	3540
MR		3325
MS		3135
M2		3760
M3		3635
M4		3385

TABLE 3 (CON'T) – REFRIGERANT CHARGE

Evap Code	Cond Code	Factory Charged Lbs.
MQ	MP, MQ, MR, MS M2, M3, M4	3394
MR		3194
MS		2988
M2		3665
M3		3496
M4		3236
NQ	NP, NQ, NR, NS N2, N3, N4	3953
NR		3689
NS		3451
N2		4233
N3		4038
N4		3738
PQ	PQ, PR, PS P2, P3, P4	3770
PR		3561
PS		3452
P2		3862
P3		3677
P4		3508
QQ	QQ, QR, QS Q2, Q3, Q4	4214
QR		3980
QS		3858
QT		4215
QV		3939
Q2		4265
Q3		4060
Q4		3874
RP	RQ, RR, RS R2, R3, R4	4775
RQ		4673
RR		4429
RS		4359
RT		4046
RV		4183
R2		4581
R3		4539
R4		4364
R5		4418
R6		4137
R7		4398
SQ	SQ, SR, SS S2, S3, S4	4929
SS		4626
SV		4412
S3		4817
S5		4688
S7		4639

Evap Code	Cond Code	Factory Charged Lbs.
SQ	VP, VQ, VR, VS V2, V3, V4, V5	5467
SS		5110
SV		4950
S3		5300
S5		5171
S7		5177
WP	WQ, WR, WS W1, W2, W3, W4	7871
WR		7383
WT		6895
W1		8002
W2		7654
W4		7378
W6		7089
XQ	TP, TQ, TR, TS, T2, T3, T4, T5	5191
XR		4890
XS		4684
X2		5338
X3		4987
X4		4727
XQ	XQ, XR, XS X2, X3, X4	5689
XR		5419
XS		5212
X2		5866
X3		5515
X4		5255
ZQ	ZQ, ZR, ZS Z1, Z2, Z3, Z4	6475
ZR		6137
ZS		5911
Z1		6984
Z2		6627
Z3		6243
Z4		5958
ZQ	YW, YX Y8, Y9	6555
ZR		6305
ZS		6060
Z1		7175
Z2		6805
Z3		6400
Z4		6150

CHECKING THE REFRIGERANT CHARGE DURING UNIT SHUTDOWN

The refrigerant charge is specified for each chiller model (See Table 3). Charge the correct amount of refrigerant and record the level in the evaporator sight glass.

The refrigerant charge should always be checked and trimmed when the system is shut down.

The refrigerant charge level must be checked after the pressure and temperature have equalized between the condenser and evaporator. This would be expected to be 4 hours or more after the compressor and water pumps are stopped. The level should visible in the sight glass.

Charge the refrigerant in accordance with the method shown under the "Refrigerant Charging", above. The refrigerant level should be observed and the level recorded after initial charging.

HANDLING REFRIGERANT FOR DISMANTLING AND REPAIRS

If it becomes necessary to open any part of the refrigerant system for repairs, it will be necessary to remove the charge before opening any part of the unit. If the chiller is equipped with optional valves, the refrigerant can be isolated in either the condenser or evaporator / compressor while making any necessary repairs.

MEGGING THE MOTOR

While the main disconnect switch and compressor motor starter are open, meg the motor as follows:

1. Using a megohm meter (megger), meg between phases and each phase to ground (See Fig. 11); these readings are to be interpreted using the graph shown in Fig. 12.
2. If readings fall below shaded area, remove external leads from motor and repeat test.

NOTE

Motor is to be megged with the starter at ambient temperature after 24 hours of idle standby.

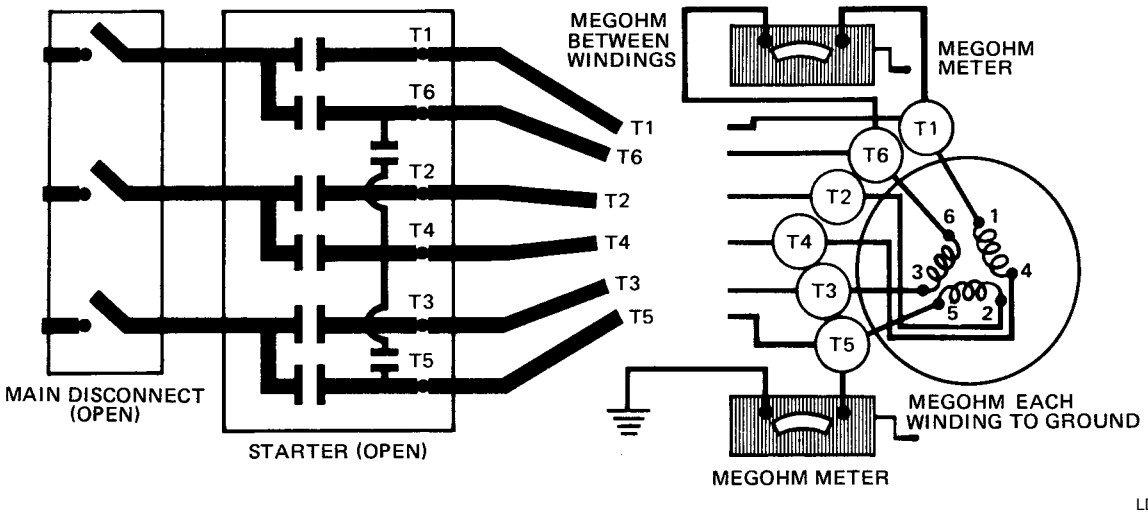


FIG. 11 – DIAGRAM, MEGGING MOTOR WINDINGS

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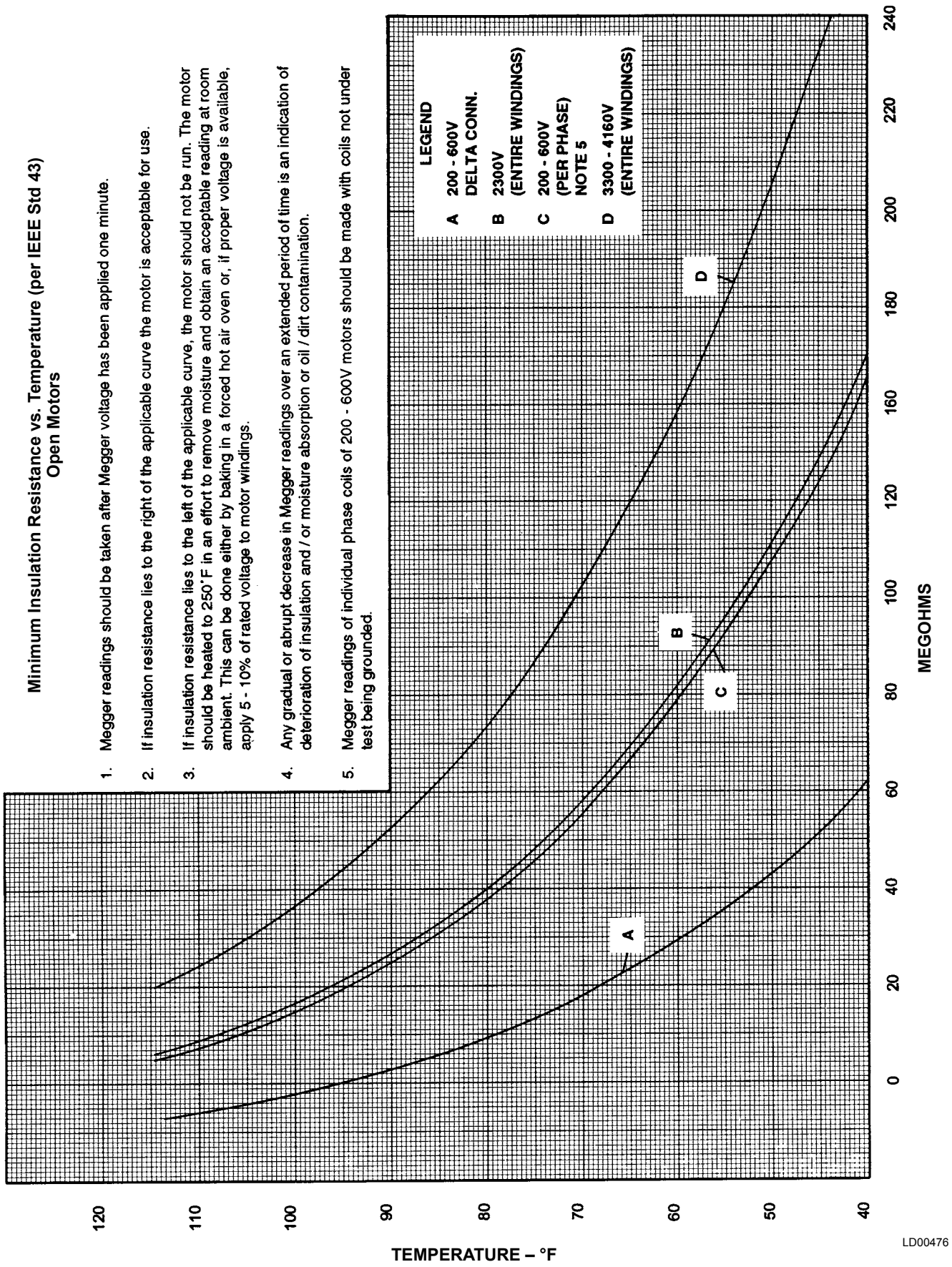


FIG. 12 – MOTOR STARTER TEMPERATURE AND INSULATION RESISTANCES

CONDENSERS AND EVAPORATORS

GENERAL

Maintenance of condenser and evaporator shells is important to provide trouble free operation of the chiller. The water side of the tubes in the shell must be kept clean and free from scale. Proper maintenance such as tube cleaning, and testing for leaks, is covered on the following pages.

CHEMICAL WATER TREATMENT

Since the mineral content of the water circulated through evaporators and condensers varies with almost every source of supply, it is possible that the water being used may corrode the tubes or deposit heat resistant scale in them. Reliable water treatment companies are available in most larger cities to supply a water treating process which will greatly reduce the corrosive and scale forming properties of almost any type of water.

As a preventive measure against scale and corrosion and to prolong the life of evaporator and condenser tubes, a chemical analysis of the water should be made preferably before the system is installed. A reliable water treatment company can be consulted to determine whether water treatment is necessary, and if so, to furnish the proper treatment for the particular water condition.

CLEANING EVAPORATOR AND CONDENSER TUBES

Evaporator

It is difficult to determine by any particular test whether possible lack of performance of the water evaporator is due to fouled tubes alone or due to a combination of troubles. Trouble which may be due to fouled tubes is indicated when, over a period of time, the cooling capacity decreases and the split (temperature difference between water leaving the evaporator and the refrigerant temperature in the evaporator) increases. A gradual drop-off in cooling capacity can also be caused by a gradual leak of refrigerant from the system or by a combination of fouled tubes and shortage of refrigerant charge. An excessive quantity of oil in the evaporator can also contribute to erratic performance.

Condenser

In a condenser, trouble due to fouled tubes is usually indicated by a steady rise in head pressure, over a period of time, accompanied by a steady rise in condensing temperature, and noisy operation. These symptoms may also be due to foul gas buildup. Purging will remove the foul gas revealing the effect of fouling.

TUBE FOULING

Fouling of the tubes can be due to deposits of two types as follows:

1. **Rust or sludge** – which finds its way into the tubes and accumulates there. This material usually does not build up on the inner tube surfaces as scale, but does interfere with the heat transfer. Rust or sludge can generally be removed from the tubes by a thorough brushing process.
2. **Scale** – due to mineral deposits. These deposits, even though very thin and scarcely detectable upon physical inspection, are highly resistant to heat transfer. They can be removed most effectively by circulating an acid solution through the tubes.

TUBE CLEANING PROCEDURES

Brush Cleaning of Tubes

If the tube consists of dirt and sludge, it can usually be removed by means of the brushing process. Drain the water sides of the circuit to be cleaned (cooling water or chilled water) remove the heads and thoroughly clean each tube with a soft bristle bronze or nylon brush. **DO NOT USE A STEEL BRISTLE BRUSH**. A steel brush may damage the tubes.

Improved results can be obtained by admitting water into the tube during the cleaning process. This can be done by mounting the brush on a suitable length of 1/8" pipe with a few small holes at the brush end and connecting the other end by means of a hose to the water supply.

The tubes should always be brush cleaned before acid cleaning.

ACID CLEANING OF TUBES

If the tubes are fouled with a hard scale deposit, they may require acid cleaning. It is important that before acid cleaning, the tubes be cleaned by the brushing process described above. If the relatively loose foreign material is removed before the acid cleaning, the acid solution will have less material to dissolve and flush from the tubes with the result that a more satisfactory cleaning job will be accomplished with a probable saving of time.



Acid cleaning should only be performed by an expert. Please consult your local water treatment representative for assistance in removing scale buildup and preventative maintenance programs to eliminate future problems.

COMMERCIAL ACID CLEANING

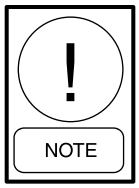
In many major cities, commercial organizations now offer a specialized service of acid cleaning evaporators and condensers. If acid cleaning is required, YORK recommends the use of this type of organization. The Dow Industries Service Division of the Dow Chemical Company, Tulsa, Oklahoma, with branches in principal cities is one of the most reliable of these companies.

TESTING FOR EVAPORATOR AND CONDENSER TUBE LEAKS

Evaporator and condenser tube leaks in R-134a systems may result in refrigerant leaking into the water circuit, or water leaking into the shell depending on the pressure levels. If refrigerant is leaking into the water, it can be detected at the liquid head vents after a period of shutdown. If water is leaking into the refrigerant, system capacity and efficiency will drop off sharply. If a tube is leaking and water has entered the system, the evaporator and condenser should be valved off from the rest of the water circuit and drained immediately to prevent severe rusting and corrosion. The refrigerant system should then be drained and purged with dry nitrogen to prevent severe rusting and corrosion. If a tube leak is indicated, the exact location of the leak may be determined as follows:

1. Remove the heads and listen at each section of tubes for a hissing sound that would indicate gas leakage. This will assist in locating the section of tubes to be further investigated. If the probable location of the leaky tubes has been determined, treat that section in the following manner (if the location is not definite, all the tubes will require investigations).

2. Wash off both tube heads and the ends of all tubes with water.



Do not use carbon tetrachloride for this purpose since its fumes give the same flame discoloration that the refrigerant does.

3. With nitrogen or dry air, blow out the tubes to clear them of traces of refrigerant laden moisture from the circulation water. As soon as the tubes are clear, a cork should be driven into each end of the tube. Pressurize the dry system with 50 to 100 PSIG (345 to 690 kPa) of nitrogen. Repeat this with all of the other tubes in the suspected section or, if necessary, with all the tubes in the evaporator or condenser. Allow the evaporator or condenser to remain corked up to 12 to 24 hours before proceeding. Depending upon the amount of leakage, the corks may blow from the end of a tube, indicating the location of the leakage. If not, if will be necessary to make a very thorough test with the leak detector.

4. After the tubes have been corked for 12 to 24 hours, it is recommended that two men working at both ends of the evaporator carefully test each tube – one man removing corks at one end and the other at the opposite end to remove corks and handle the leak detector. Start with the top row of tubes in the section being investigated. Remove the corks at the ends of one tube simultaneously and insert the exploring tube for 5 seconds – this should be long enough to draw into the detector any refrigerant gas that might have leaked through the tube walls. A fan placed at the end of the evaporator opposite the detector will assure that any leakage will travel through the tube to the detector.

5. Mark any leaking tubes for later identification.

6. If any of the tube sheet joints are leaking, the leak should be indicated by the detector. If a tube sheet leak is suspected, its exact location may be found by using a soap solution. A continuous buildup of bubbles around a tube indicates a tube sheet leak.

COMPRESSOR

Maintenance for the compressor assembly consists of checking the operation of the oil return system and changing the dehydrator, checking and changing the oil, checking and changing the oil filters, checking the operation of the oil heater, checking the operation of the oil pump, and observing the operation of the compressor.

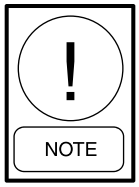
Internal wearing of compressor parts could be a serious problem caused by improper lubrication, brought about by restricted oil lines, passages, or dirty oil filters. If the unit is shutting down on (HOT) High Oil Temperature or Low Oil Pressure (OP), change the oil filter element. Examine the oil filter element for the presence of aluminum particles. Aluminum gas seal rings can contact the impeller and account for some aluminum particles to accumulate in the oil filter, especially during the initial start up and first several months of operation. However, if aluminum particles continue to accumulate and the same conditions continue to stop the unit operation after a new filter is installed, notify the nearest Johnson Controls office to request the presence of a Johnson controls Service Technician.

ELECTRICAL CONTROLS

For information covering the OptiView™ Control Center operation, refer to Form 160.54-O1.

SECTION 7
PREVENTIVE MAINTENANCE

It is the responsibility of the owner to provide the necessary daily, monthly and yearly maintenance requirements of the system.



IMPORTANT – If a unit failure occurs due to improper maintenance during the warranty period; Johnson Controls will not be liable for costs incurred to return the system to satisfactory operation.

In any operating system it is most important to provide a planned maintenance and inspection of its functioning parts to keep it operating at its peak efficiency. Therefore, the following maintenance should be performed when prescribed.

COMPRESSOR

1. **Oil Filter** – When oil pump VSD frequency increases to 55 hz to maintain target oil pressure.

   When the oil filter is changed, it should be inspected thoroughly for any aluminum particles which would indicate possible bearing wear. If aluminum particles are found this should be brought to the attention of the nearest Johnson Controls office for their further investigation and recommendations.
2. **Oil Changing** – The oil in the compressor must be changed annually or earlier if it becomes dark or cloudy. However, quarterly oil analysis can eliminate the need for an annual change provided the analysis indicates there is no problem with the oil.

COMPRESSOR MOTOR

1. Check motor mounting screws frequently to insure tightness.
2. Meg motor windings annually to check for deterioration of windings.

GREASED BEARINGS

Motor Operation and Maintenance manuals are supplied with the chillers providing maintenance schedules and instructions for the specific motors. The following are lubrication schedules for the most common motors:

RAM Motor Lubrication

Frame 143T thru 256T are furnished with double sealed ball bearings, pre-lubricated prior to installation. Grease fittings are not supplied and bearings are designed for long life under standard conditions.

Frames 284T thru 587UZ are furnished with double shielded or open ball or roller bearings. It is necessary to re-lubricate anti-friction bearings periodically. (See Table 4)

TABLE 4 – BEARING LUBRICATION

Frame Size	Standard 8hr/Day	Continuous 24hr Day	Grease Qty. oz.
143T-256T	*7 Years	*3 Years	*1
284TS - 286TS	210 Days	70 Days	1
324TS - 587USS	150 Days	50 Days	2

* - On frame sizes 143T - 256T, changing bearings at these intervals is recommended. However, removing the seal, cleaning and refilling the bearing and the cavity with recommended grease can re-lubricate these bearings.

Recommended greases for standard applications:

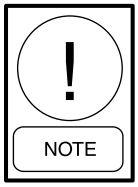
OPERATING AMBIENT TEMP. -30ºC to 50ºC	
Chevron SRI	(Chevron)
Exxon Unirex #2	(Exxon Corp.)
Exxon Polyrex	(Exxon Corp.)
Shell Dolum R	(Shell Oil Co.)

Westinghouse Motor Lubrication:

- Re-greasing should occur at 1000 operating hour intervals.
- Westinghouse recommends using Westinghouse Grease No. 53701.
- Motors with shaft diameters less than 2 3/8 inch require 1 oz of grease per bearing while motors with shaft diameters between 2 3/8 and 3 inches require 1.5 oz.



Mixing different greases is not recommended



Additional information on motor lubrication and other service issues can be found in the A-C Motors Instruction Manual.

LEAK TESTING

The unit should be leak tested monthly. Any leaks found must be repaired immediately.

EVAPORATOR AND CONDENSER

The major portion of maintenance on the condenser and evaporator will deal with the maintaining the water side of the condenser and evaporator in a clean condition.

The use of untreated water in cooling towers, closed water systems, etc. frequently results in one or more of the following:

1. Scale Formation.
2. Corrosion or Rusting.
3. Slime and Algae Formation.

It is therefore to the benefit of the user to provide for proper water treatment to provide for a longer and more economical life of the equipment. The following recommendation should be followed in determining the condition of the water side of the condenser and evaporator tubes.

1. The condenser tubes should be cleaned annually or earlier if conditions warrant. If the temperature difference between the water off the condenser and the condenser liquid temperature is more than 4°F (2°C) greater than the difference recorded on a new unit, it is a good indication that the condenser tubes require cleaning. Refer to the Maintenance section of this manual for condenser tube cleaning instructions.
2. The evaporator tubes under normal circumstances will not require cleaning. If the temperature difference between the refrigerant and the chilled water increases slowly over the operating season, it is an indication that the evaporator tubes may be fouling or that there may be a water bypass in the water box requiring gasket replacement or refrigerant may have leaked from the chiller.
3. Heat recovery condenser tubes should be evaluated similiar to evaporator tubes when the heating circuit is a treated, closed loop. Fouling could be detected as ability to meet heat load requirements decreases.

OIL RETURN SYSTEM

1. Change the dehydrator in the oil return system semi-annually or earlier if the oil return system fails to operate.
2. When the dehydrator is changed, the nozzle of the eductor should be checked for any foreign particles that may be obstructing the jet.

ELECTRICAL CONTROLS

1. All electrical controls should be inspected for obvious malfunctions.
2. It is important that the factory settings of controls (operation and safety) not be changed. If the settings are changed without Johnson Controls approval, the warranty will be jeopardized.

SI METRIC CONVERSION

The following factors can be used to convert from English to the most common SI Metric values.

MEASUREMENT	MULTIPLY THIS *ENGLISH* VALUE	BY	TO OBTAIN THIS *METRIC* VALUE
CAPACITY	TONS REFRIGERANT EFFECT (ton)	3.516	KILOWATTS (kW)
POWER	KILOWATTS (kW)	NO CHANGE	KILOWATTS (kW)
	HORSEPOWER (hp)	0.7457	KILOWATTS (kW)
FLOW RATE	GALLONS / MINUTE (gpm)	0.0631	LITERS / SECOND (L/s)
LENGTH	FEET (ft)	304.8	MILLIMETERS (mm)
	INCHES (in)	25.4	MILLIMETERS (mm)
WEIGHT	POUNDS (lb)	0.4536	KILOGRAMS (kg)
VELOCITY	FEET / SECOND (fps)	0.3048	METERS / SECOND (m/s)
PRESSURE DROP	FEET OF WATER (ft)	2.989	KILOPASCALS (k Pa)
	POUNDS / SQ. INCH (psi)	6.895	KILOPASCALS (k Pa)

TEMPERATURE

To convert degrees Fahrenheit (°F) to degrees Celsius (°C), subtract 32° and multiply by 5/9 or 0.5556.

To convert a temperature range (i.e., 10°F or 12°F chilled water range) from Fahrenheit to Celsius, multiply by 5/9 or 0.5556.

MAINTENANCE REQUIREMENTS FOR YORK YK CHILLERS

PROCEDURE	DAILY	WEEKLY	MONTHLY	YEARLY	OTHER
Record operating conditions (on applicable Log Form)	X				
Check oil levels	X				
Check refrigerant levels		X			
Check oil return system operation			X		
Check operation of motor starter			X		
Check sump heater and thermostat operation			X		
Check three-phase voltage and current balance			X		
Verify proper operation/setting/calibration of safety controls[1]			X		
Verify condenser and evaporator water flows			X		
Leak check and repair leaks as needed[1]			X		
Check and tighten all electrical connections				X	
Megohm motor windings				X	
Replace oil filter and oil return filter/driers				X	
Clean or backflush heat exchanger (VSD, SSS Applications)				X	
Replace starter coolant (VSD, SSS Applications)				X	
Replace or clean starter air filters if applicable				X[2]	
Perform oil analysis on compressor lube oil[1]				X	
Perform refrigeration analysis[1]				X	
Perform vibration analysis				X	
Clean tubes				X[2]	
Perform Eddy current testing and inspect tubes					2 - 5 Years
Lubricate motor	Refer to motor manufacturer's recommentations				

For operating and maintenance requirements listed above, refer to appropriate service literature, or contact your local YORK Service Office.

[1] This procedure must be performed at the specified time interval by an Industry Certified Technician who has been trained and qualified to work on this type of YORK equipment. A record of this procedure being successfully carried out must be maintained on file by the equipment owner should proof of adequate maintenance be required at a later date for warranty validation purposes.

[2] More frequent service may be required depending on local operating conditions.

Form 160.54-MR1 (801)

NOTES

NOTES



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